

Safety data sheet

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BASF Safety data sheet according to 91/155/EEC

Date / Revised: 04.02.2005

Version: 3.0

Product: **Formic acid 94%**

(30192634/SDS_GEN_EU/EN)

Date of print 10.03.2006

1. Substance/preparation and company identification

Formic acid 94%

Use: Intermediate

Company:

BASF Aktiengesellschaft

67056 Ludwigshafen

Operating Division Intermediates

Telephone: 0621-60-42259

Telefax number: 0621-60-49134

E-mail address: product-safety.intermediates@basf.com

Emergency information:

Telephone: +49-1802273 112

Telefax number: 0621-60-92664

2. Composition/information on ingredients

Chemical nature

| formic acid (Content (W/W): 94 %)

CAS Number: 64-18-6

EC-Number: 200-579-1

INDEX-Number: 607-001-00-0

| water (Content (W/W): 6 %)

CAS Number: 7732-18-5

EC-Number: 231-791-2

Hazardous ingredients

| formic acid

Content (W/W): 94 %

CAS Number: 64-18-6

EC-Number: 200-579-1

INDEX-Number: 607-001-00-0

Hazard symbol(s): C

| R-phrases: 35

The wording of the hazard symbols and R-phrases is specified in chapter 16 if dangerous ingredients are mentioned.

3. Hazard identification

Causes severe burns.

4. First-aid measures

General advice:

Immediately remove contaminated clothing. If danger of loss of consciousness, place patient in recovery position and transport accordingly. Apply artificial respiration if necessary. First aid personnel should pay attention to their own safety.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Rinse mouth immediately and then drink plenty of water, seek medical attention.

5. Fire-fighting measures

Suitable extinguishing media:

water, dry extinguishing media, alcohol-resistant foam, carbon dioxide

Specific hazards:

| carbon monoxide

The substances/groups of substances mentioned can be released if the product is involved in a fire.

Special protective equipment:

Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:

Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental release measures

Personal precautions:

Breathing protection required. Avoid contact with the skin, eyes and clothing.

Environmental precautions:

Do not empty into drains.

Methods for cleaning up or taking up:

For large amounts: Pump off product.

For residues: Pick up with suitable absorbent material (e.g. acid binder). Dispose of absorbed material in accordance with regulations.

7. Handling and storage

Handling

Ensure thorough ventilation of stores and work areas. Sealed containers should be protected against heat as this results in pressure build-up.

Protection against fire and explosion:

Sources of ignition should be kept well clear.

Storage

Segregate from alkalies and alkalizing substances.

Further information on storage conditions: Danger of bursting when sealed gastight.

Storage stability:

Storage temperature: < 30 °C

Storage duration: 24 Months

8. Exposure controls and personal protection

Personal protective equipment

Respiratory protection:

Suitable respiratory protection for lower concentrations or short-term effect: Gas filter EN141 Type A for gases/vapours of organic compounds (boiling point >65 °C). Suitable respiratory protection for higher concentrations or long-term effect: Self-contained breathing apparatus.

Hand protection:

Chemical resistant protective gloves (EN 374)

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374):

chloroprene rubber (CR) - 0.5 mm coating thickness

butyl rubber (butyl) - 0.7 mm coating thickness

Manufacturer's directions for use should be observed because of great diversity of types.

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Supplementary note: The specifications are based on own tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined in accordance with EN 374.

Eye protection:

Tightly fitting safety goggles (cage goggles) (EN 166) and face shield.

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to DIN-EN 465).

General safety and hygiene measures:

Avoid contact with the skin, eyes and clothing. Avoid inhalation of vapour.

9. Physical and chemical properties

Form:	liquid	
Colour:	colourless to yellow	
Odour:	pungent odour	
pH value:	2.2 (10 g/l, 20 °C)	
Melting point:	-2 °C	
Boiling point:	103 °C	
Flash point:	56 °C	(DIN 51755)
Lower explosion limit:	13.9 %(V)	
Upper explosion limit:	38.1 %(V)	
Ignition temperature:	480 °C	(DIN 51794)
Vapour pressure:	36 mbar (20 °C) 148 mbar (50 °C)	
Density:	1.21 g/cm ³ (20 °C)	
Miscibility with water:	miscible in all proportions	
Solubility (qualitative) solvent(s):	organic solvents miscible	
Partitioning coefficient n-octanol/water (log Pow):	-0.54	
Viscosity, dynamic:	1.6 mPa.s (20 °C)	

10. Stability and reactivity

Hazardous reactions:

Exothermic reaction. Reacts with alkalies. Reacts with amines.

Possible thermal decomposition products:

| carbon monoxide

11. Toxicological information

LD50/oral/rat: 730 mg/kg

The data on toxicology refer to the active ingredient.

LC50/by inhalation/rat: 7.4 mg/l / 4 h

The data on toxicology refer to the active ingredient.

by inhalation/rat: / 3 min

Inhalation-risk test (IRT): Mortality within 10 minutes as shown in animal studies. The inhalation of a highly saturated vapor-air mixture represents a severe hazard.

Primary skin irritation/rabbit: Corrosive.

Primary irritations of the mucous membrane:

As the product corrodes the skin, it can be expected to have a similar effect on the eyes also.

Information on: formic acid

Sensitization/Buehler test/guinea pig: Skin sensitizing effects were not observed in animal studies. (OECD Guideline 406)

Further information:

The substance was not mutagenic in bacteria.

12. Ecological information

Ecotoxicity

Toxicity to fish:

Leuciscus idus/LC50 (96 h): > 46 - < 100 mg/l

Aquatic invertebrates:

| Daphnia magna/LC50 (48 h): 32.19 mg/l

Aquatic plants:

Scenedesmus subspicatus/EC50 (72 h): 26.9 mg/l

Microorganisms/Effect on activated sludge:

Pseudomonas putida/EC50 (17 h): 46.7 mg/l

DIN/EN/ISO 8192-OECD 209-88/302/EEC, P. C

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activated sludge/EC20: > 1,000 mg/l

The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Persistence and degradability

Elimination information

Test method: OECD 301E; 84/449/EEC, C.3
Method of analysis: DOC reduction
Degree of elimination: > 90 %
Evaluation: Readily biodegradable.

Chemical oxygen demand (COD): 348 mg/g

Biochemical oxygen demand (BOD) Incubation period 5 d: 86 mg/g

Bioaccumulation potential

Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

13. Disposal considerations

Incinerate in suitable incineration plant, observing local authority regulations.

Contaminated packaging:

Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

14. Transport information

Land transport

ADR	: Class	8
	Packaging group	II
	UN-number	1779
	Designation of goods	FORMIC ACID

RID	: Class	8
	Packaging group	II
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Inland waterway transport

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ADNR	: Class	8
	Packaging group	II
	UN-number	1779
	Designation of goods	FORMIC ACID

Sea transport

IMDG/GGVSee	: Class	8
	Packaging group	II
	UN-number	1779
	Marine pollutant	NO
	Exact technical name	FORMIC ACID

Air transport

ICAO/IATA	: Class	8
	Packaging group	II
	UN-number	1779
	Exact technical name	FORMIC ACID

15. Regulatory information**Regulations of the European union (Labelling) / National legislation/Regulations**

EC-Number: 200-579-1

Directive 1999/45/EC ('Preparation Directive'):

Hazard symbol(s)

C	Corrosive.
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R-phrases)

R35	Causes severe burns.
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S-phrases)

S23.5	Do not breathe vapour.
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S26	In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
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S45	In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).
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Hazard determinant component(s) for labelling: FORMIC ACID

Other regulations

16. Other information

| Recommended use: dyes, pesticides, industrial chemicals, initial product for chemical syntheses

flue gas desulphurization rubber industry textile industry leather industry

Full text of hazard symbols and R-phrases if mentioned as hazardous components in chapter 2:

C	Corrosive.
35	Causes severe burns.

Vertical lines in the left hand margin indicate an amendment from the previous version.

The information contained herein is based on the present state of our knowledge and does not therefore guarantee certain properties. Recipients of our product must take responsibility for observing existing laws and regulations.

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